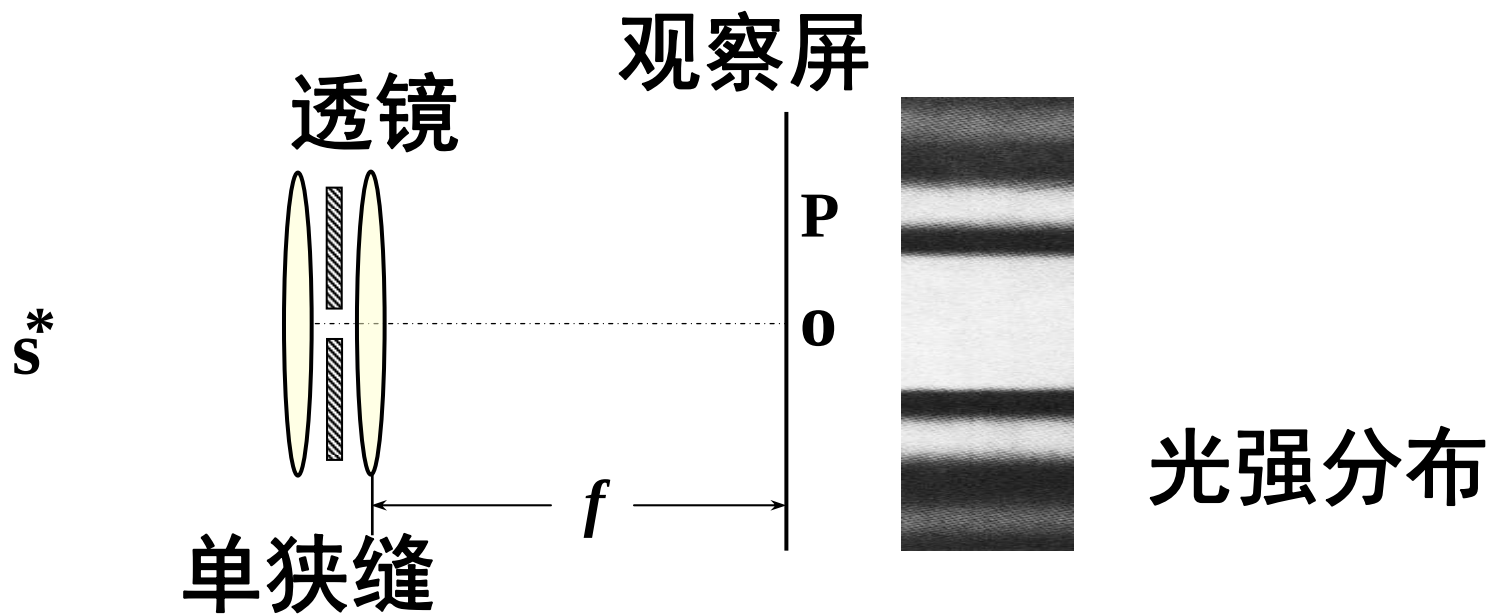
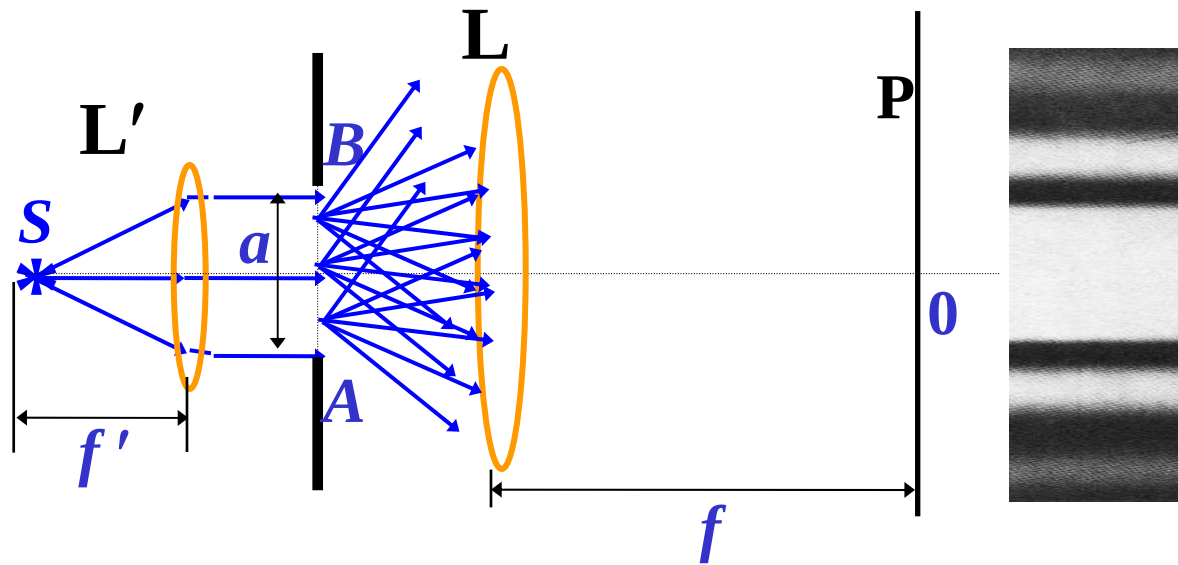


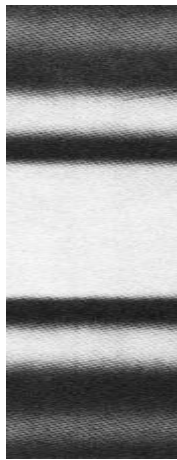
3.5 夫琅和费的单缝衍射

3.5.1 实验装置及光路示意图



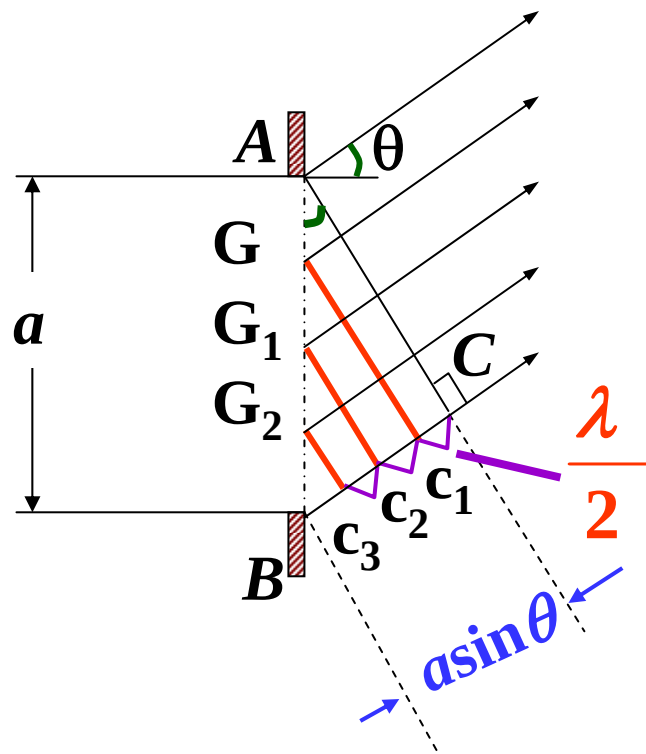
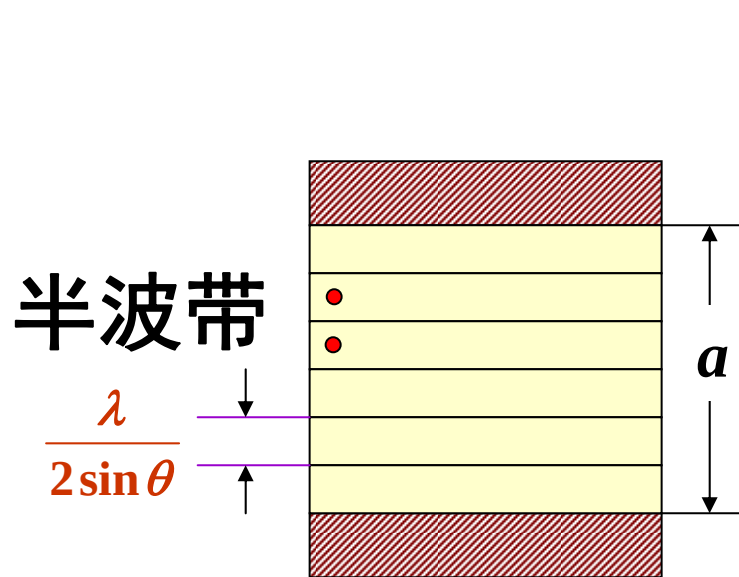
光路示意图





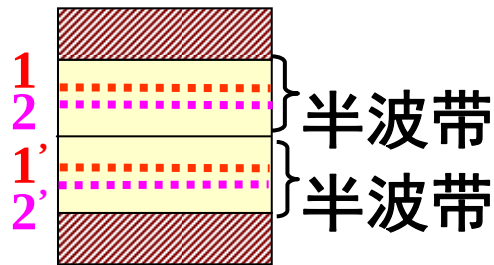
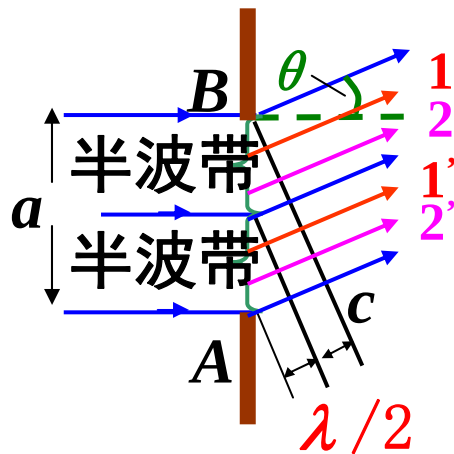
菲涅耳积分 半波带法

3.5.2 用菲涅耳半波带分析夫琅和费单缝衍射图样



分成偶数个半波带 --- 暗纹

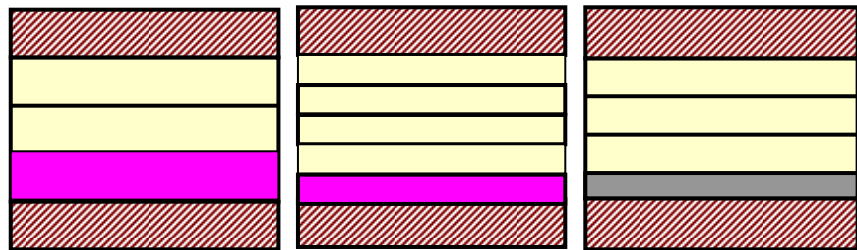
$$a \sin \theta = \pm 2k \frac{\lambda}{2} = \pm k \lambda \quad k=1, 2, 3$$



分成奇数个半波带 --- 明纹

$$a \sin \theta = \pm (2k + 1) \frac{\lambda}{2}$$

$$k=1, 2, 3$$

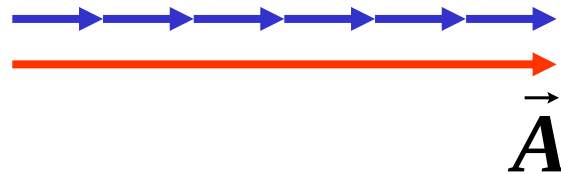
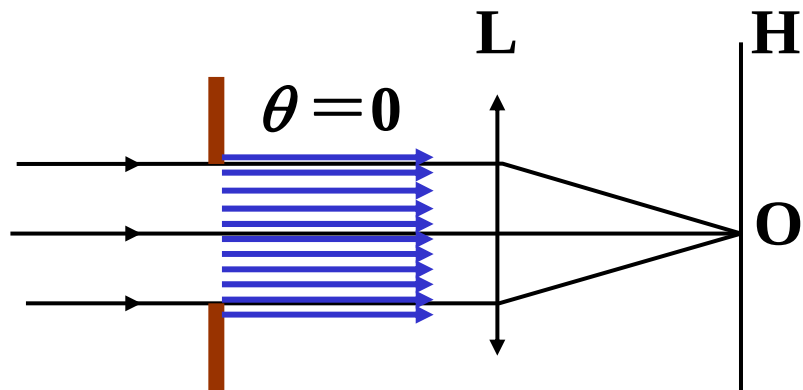


$\theta \uparrow \rightarrow$ 波带数 $\uparrow \rightarrow$ 光强 $\downarrow$

不能分成整数个半波带

光强介于最强与最暗之间

$$\theta = 0, \quad \delta = 0$$



— 中央明纹(中心)

明暗纹条件

$$a \sin \theta = 0$$

中央明纹(中心)

$$a \sin \theta = \pm(2k+1)\frac{\lambda}{2}, \quad k=1,2,3\cdots$$

明纹中心(近似)

$$a \sin \theta = \pm k\lambda, \quad k=1,2,3\cdots$$

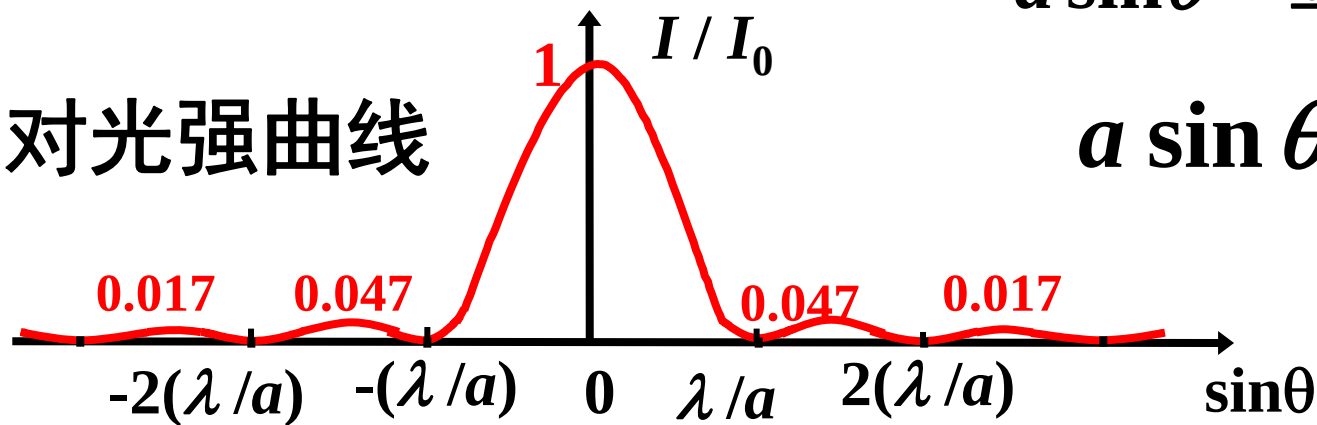
暗纹(中心)

若 $a \sin \theta$ 不是半波长的整数倍, 亮度介于最明和最暗之间

3.5.3夫琅和费单缝衍射图样的特点

光强分布

相对光强曲线



$$a \sin \theta = 0$$

$$a \sin \theta = \pm(2k+1)\frac{\lambda}{2}$$

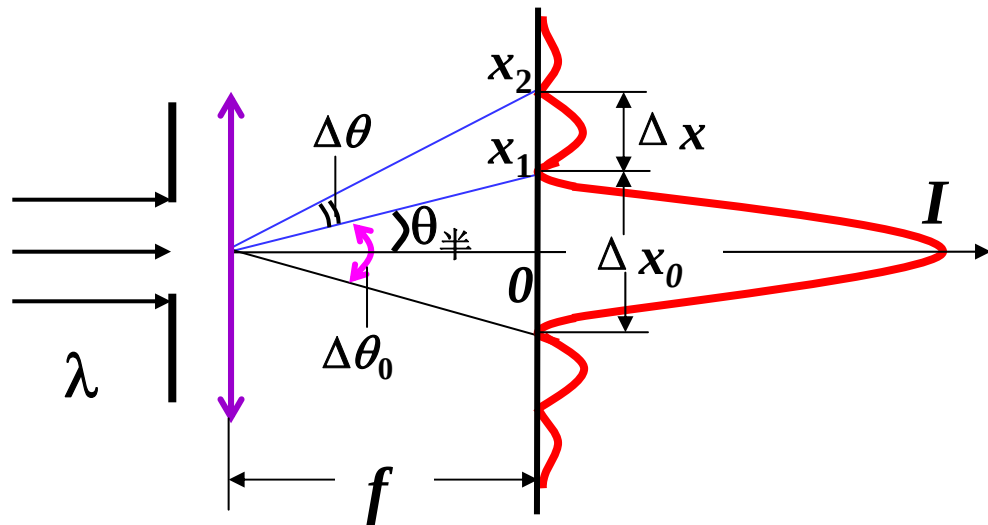
$$a \sin \theta = \pm k\lambda$$



条纹宽度

1)角宽度

某一亮纹的角宽度为该亮纹两侧两相邻暗纹中心对透镜光心所张的角度

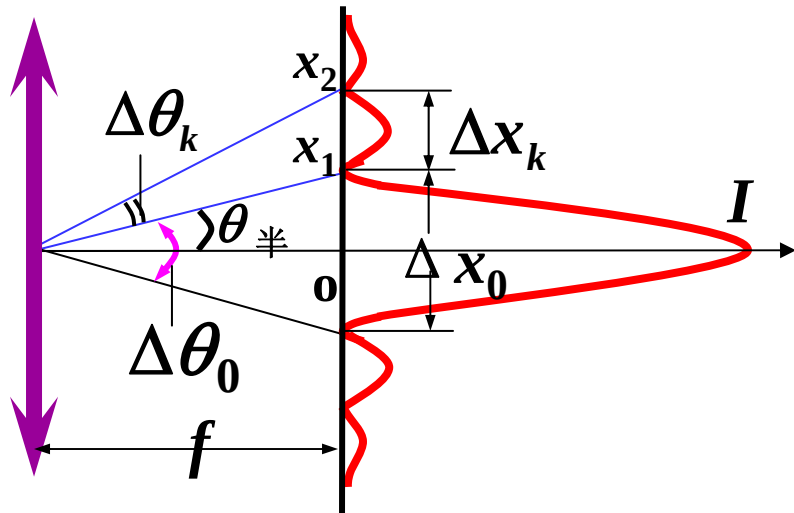


k 级明纹角宽度

$$a \sin \theta_k = \pm k\lambda \quad \sin \theta_k \approx \theta_k$$

$$\theta_k = \frac{k\lambda}{a}$$

$$\Delta\theta_k = \theta_{k+1} - \theta_k = \frac{\lambda}{a}$$



中央明纹角宽度

$$\Delta\theta_0 = \theta_{+1} - \theta_{-1} = \frac{\lambda}{a} - \left(-\frac{\lambda}{a}\right) = \frac{2\lambda}{a} \quad \Delta\theta_{\frac{\pi}{2}} = \frac{\lambda}{a}$$

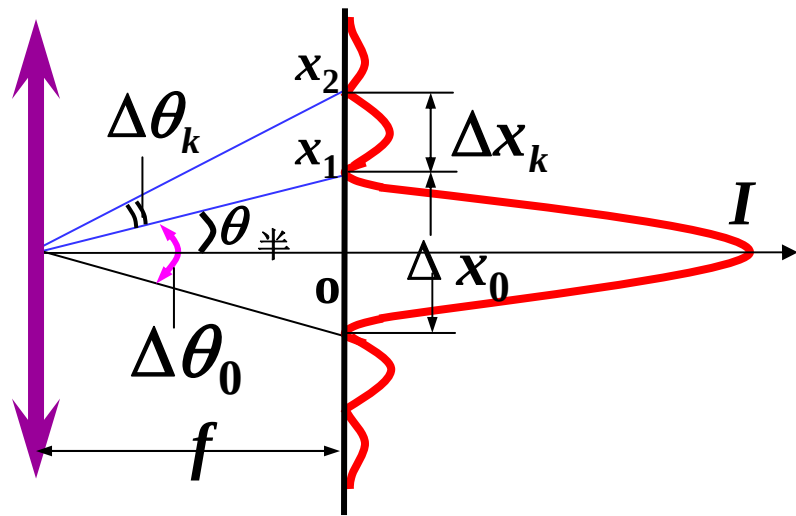
2) 亮纹的线宽度

中央亮纹

$$\begin{aligned}\Delta x_0 &= 2f \operatorname{tg} \theta_{\frac{\pi}{2}} \\ &= 2f \theta_{\frac{\pi}{2}} = 2f \frac{\lambda}{a} \propto \frac{\lambda}{a}\end{aligned}$$

其它次极大

$$\Delta x = f \Delta \theta = f \frac{\lambda}{a}$$



—衍射反比定律

讨论

$$\theta_k = \frac{k\lambda}{a} \quad \Delta\theta_{\frac{1}{2}} = \frac{\lambda}{a} \quad \Delta x = f \frac{\lambda}{a}$$

1) 波长对条纹宽度的影响

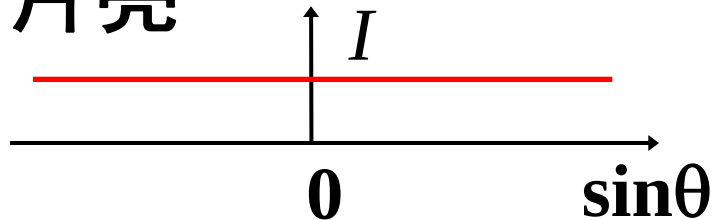
缝宽 a 一定, $\lambda \uparrow \Delta\theta \uparrow \Delta x \propto \lambda$,
波长越长, 条纹宽度越宽

2) 缝宽变化对条纹的影响

(1) $\Delta x = \frac{1}{2} \Delta x_0 = f \frac{\lambda}{a}$ 缝宽越小，条纹宽度越宽

当 $\frac{a}{\lambda} \rightarrow 0$ 时, $\Delta x \rightarrow \infty$ 屏幕是一片亮

(2) 当 $a \gg \lambda$ 时, $\frac{\lambda}{a} \rightarrow 0$ $\Delta x \rightarrow 0$



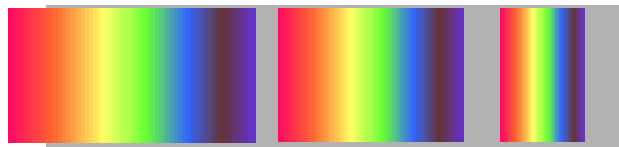
只显出单一的明条纹 -- 单缝的几何光学象

∴ 几何光学是波动光学在 $\lambda/a \rightarrow 0$ 时的极限情形

3) 白光入射单缝

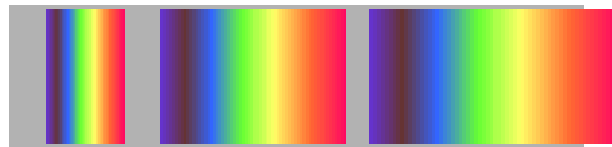
$$\theta_k = \frac{k\lambda}{a} \quad \Delta x = f \frac{\lambda}{a}$$

中央明纹



-3级光谱

-1级光谱



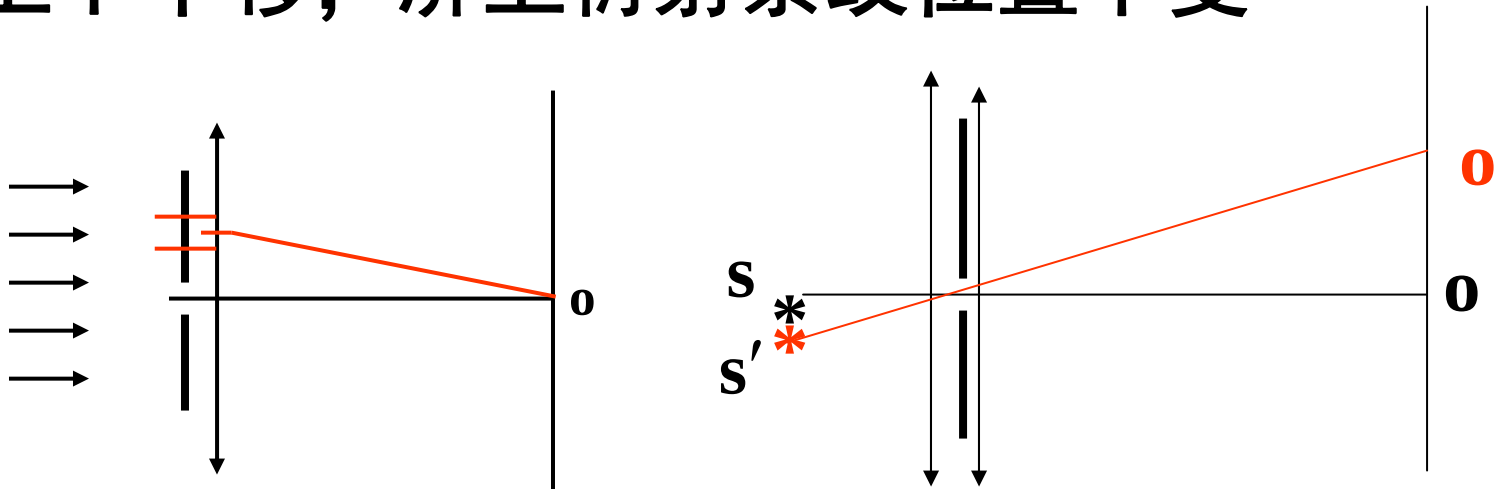
1级光谱

3级光谱

中央：白色明纹

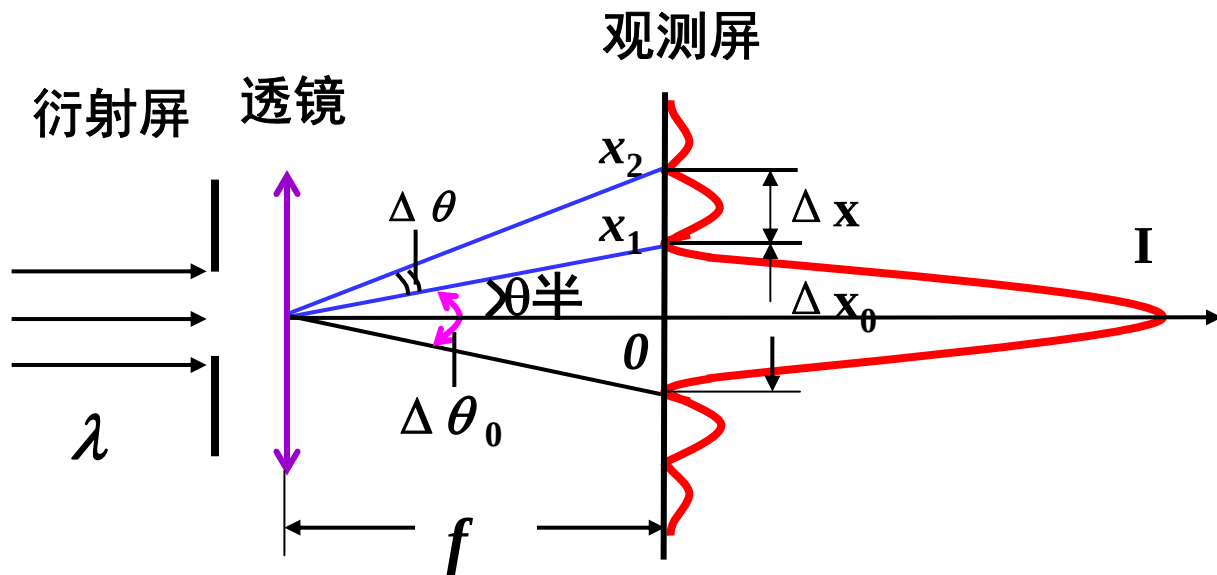
两侧：对称彩带，由紫到红

4) 单缝上下平移，屏上衍射条纹位置不变



5) 光源上下移动对条纹的影响

6) 以上明、暗纹公式只适用于平行光垂直单缝入射，入射光线倾斜，需考虑入射光的光程差



**[例]:单缝夫琅禾费衍射,已知: $a = 0.3\text{mm}$, $f = 12.62\text{cm}$,
第五级暗纹之间距离 $L = 0.24\text{cm}$**

求: 1) λ ;

2) $k=5$ 的暗纹对应的半波带数

[例]:单缝夫琅禾费衍射,已知: $a = 0.3\text{mm}$, $f = 12.62\text{cm}$,
第五级暗纹之间距离 $L = 0.24\text{cm}$

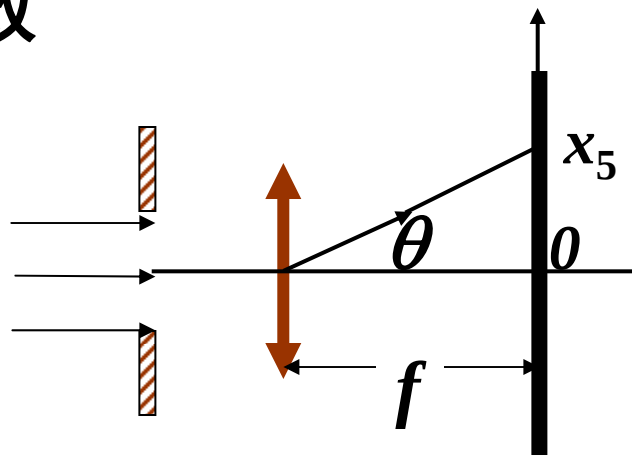
求: 1) λ ;

2) $k=5$ 的暗纹对应的半波带数

解: 1) $a \cdot \sin\theta_5 = k\lambda \quad k=5 \quad (1)$

$$L = 2x_5 \quad (2)$$

$$x_5 = f \tan\theta_5 \quad (3)$$



$$a \cdot \sin \theta_5 = k\lambda \quad k=5 \quad (1) \quad L=2x_5 \quad (2) \quad x_5 = f \tan \theta_5 \quad (3)$$

$$\text{由(1)得: } \sin \theta_5 = \frac{5\lambda}{a} \quad \text{代入(3): } x_5 = \frac{5f\lambda}{a}$$

$$L = 2x_5 = \frac{10f\lambda}{a}$$

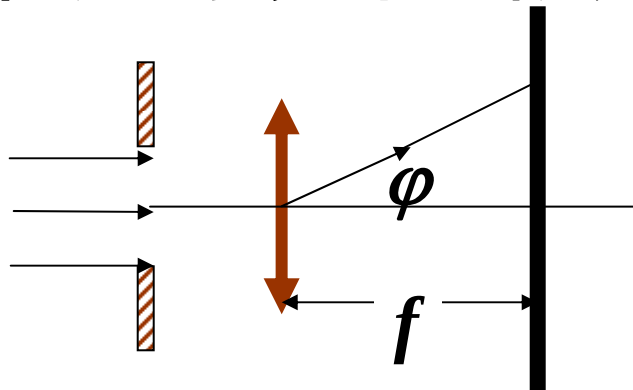
$$\lambda = \frac{aL}{10f} = \frac{0.3 \times 0.24}{10 \times 12.62} \times 10^{-7} = 5705 [\text{\AA}]$$

$$2) \quad a \cdot \sin \theta_5 = 2k \frac{\lambda}{2} \quad 2k=10 \text{ 个半波带}$$

[例]: 如图所示单缝衍射装置,缝宽为 $a = 0.5\text{mm}$,
 $f = 50\text{cm}$, 用可见光照射单缝, 在观察屏上 x
 $= 1.5\text{mm}$ 处看到明纹极大,

求:(1)入射光的波长及衍射级次

(2)单缝所在处的波阵面被分成的波带数目



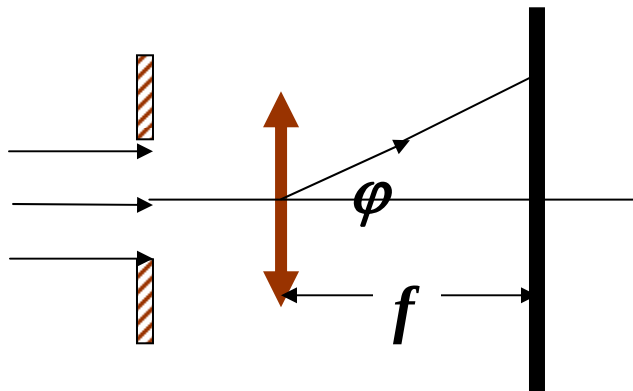
解1)

明纹 $a \sin \varphi = (2k + 1) \frac{\lambda}{2}$

$$\operatorname{tg} \varphi = \frac{x}{f} = \frac{1.5}{500} = 0.003 < 5^\circ$$

$$\sin \varphi = 0.003$$

$$\lambda = \frac{2 a \sin \varphi}{(2 k + 1)} = \frac{2 \times 0.5 \times 0.003}{(2 k + 1)} \times 10^7 = \frac{3 \times 10^4}{(2 k + 1)} (\text{\AA})$$



$$\lambda = \frac{2a \sin \varphi}{(2k + 1)} = \frac{2 \times 0.5 \times 0.003}{(2k + 1)} \times 10^7 = \frac{3 \times 10^4}{(2k + 1)} (\text{\AA})$$

$$k=1 \quad \lambda_1 = 10000(\text{\AA}) \quad \text{红外}$$

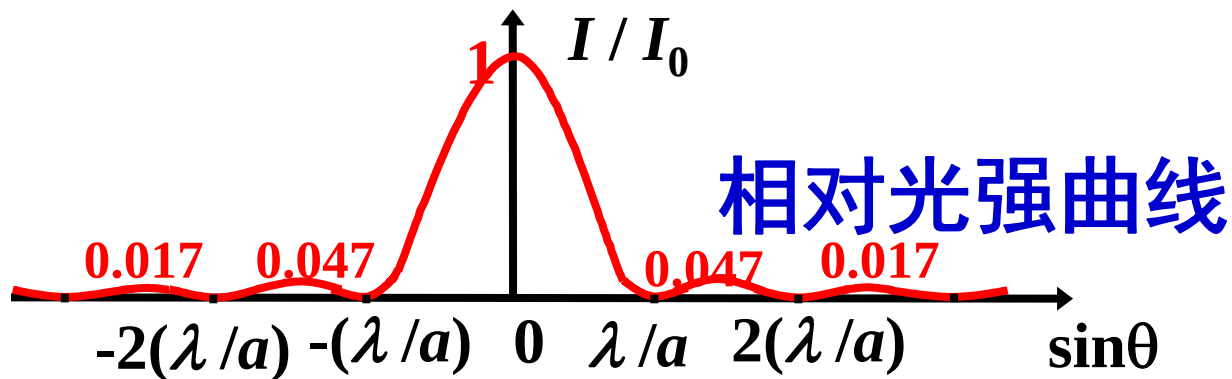
$$\left. \begin{array}{l} k=2 \quad \lambda_2 = 6000(\text{\AA}) \\ k=3 \quad \lambda_3 = 4286(\text{\AA}) \end{array} \right\} \text{可见光}$$

$$k=4 \quad \lambda_4 = 3333(\text{\AA}) \quad \text{紫外}$$

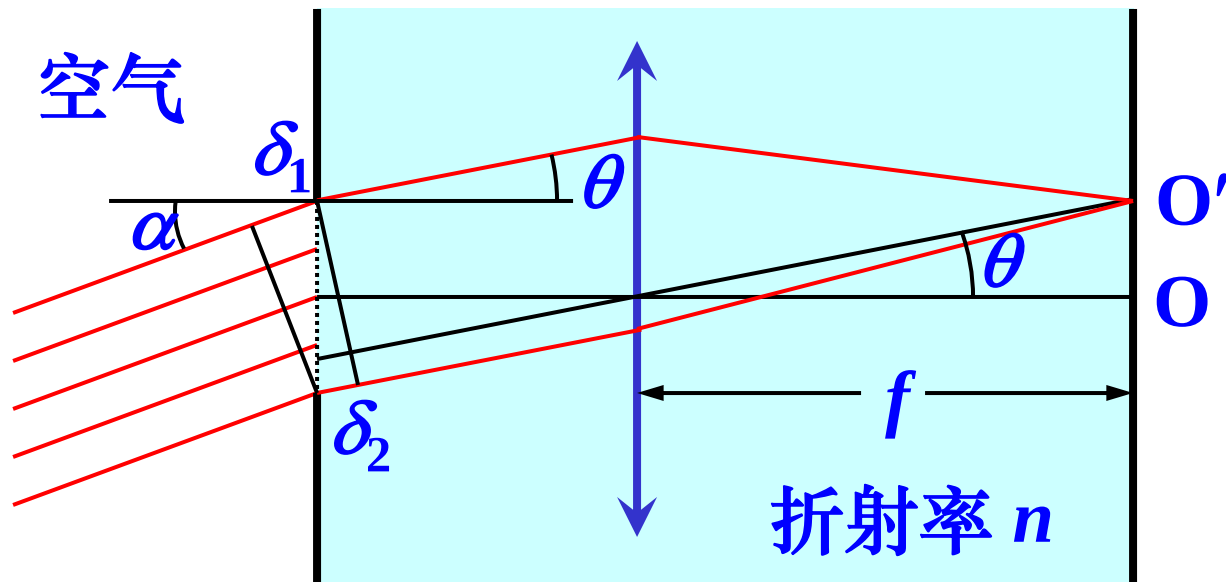
2) 波带数目

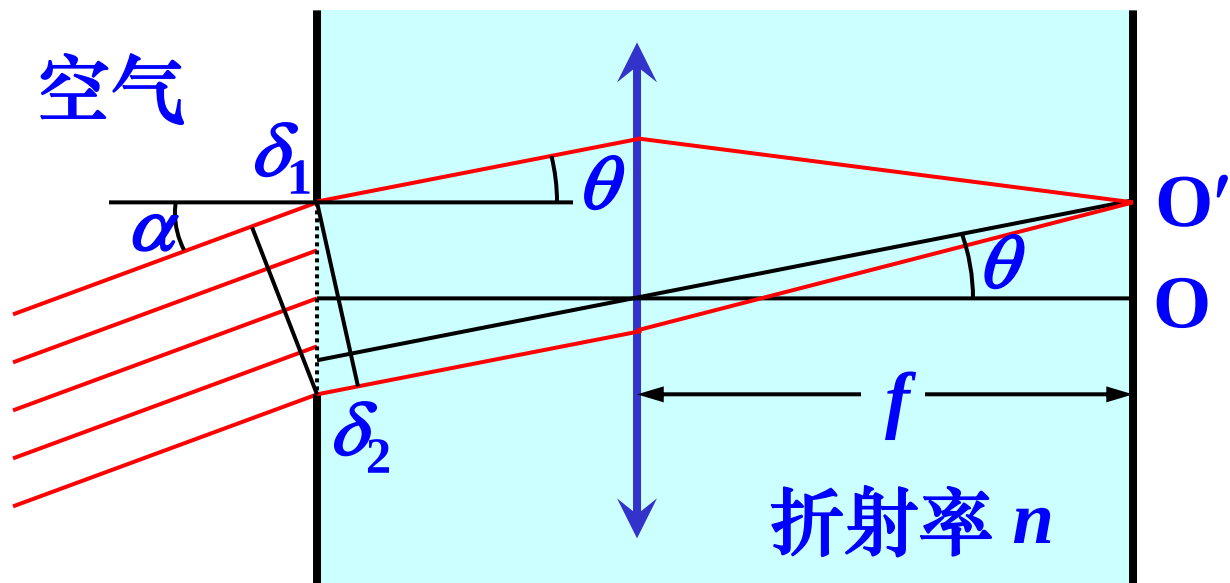
$$k=2 \quad \lambda_2 = 6000(\text{\AA}) \quad N = 2k+1 = 5$$

$$k=3 \quad \lambda_3 = 4286(\text{\AA}) \quad N = 2k+1 = 7$$



[例] 平行单色光以角 α 入射单缝，求中央明纹位置





解：平行光斜入射，中央明纹不在 O 点处
 中央明纹对应衍射光线等光程点，设其为 O'
 点，则光程差

$$\delta = \delta_1 - \delta_2 = a \sin \alpha - n a \sin \theta = 0$$

解：平行光斜入射，中央明纹不在 O 点处
中央明纹对应衍射光线等光程点，设其为 O'
点，则光程差

$$\delta = \delta_1 - \delta_2 = a \sin \alpha - n a \sin \theta = 0$$

$$\therefore \theta = \arcsin \frac{\sin \alpha}{n}, \quad |OO'| = f \operatorname{tg} \theta$$

当 $n = 1$ 时, $\theta = \alpha$